

Exercício 1

$$1. H(\vec{x}, \vec{p}) = \frac{1}{2m} [\vec{p} - e \vec{A}(\vec{x})]^2 + e \Phi(\vec{x})$$

$$\vec{\pi} \equiv \vec{p} - e \vec{A}(\vec{x})$$

$$a) \{x_i, \pi_j\} = \{x_i, p_j - e A_j\} = \{x_i, p_j\} - e \{x_i, A_j\} \quad \frac{\partial A}{\partial p} = 0$$

$$= \delta_{ij}$$

$$\{\pi_i, \pi_j\} = \{p_i - e A_i, p_j - e A_j\}$$

$$= \{p_i, p_j - e A_j\} - e \{A_i, p_j - e A_j\}$$

$$= \{p_i, p_j\} - e \{p_i, A_j\} - e \{A_i, p_j\} + e^2 \{A_i, A_j\}$$

$$= -e [\{p_i, A_j\} + \{A_i, p_j\}]$$

$$= -e \left[\cancel{\frac{\partial A_j}{\partial x_i}} - e \left[-\frac{\partial p_i}{\partial p_k} \frac{\partial A_j}{\partial x_k} + \frac{\partial A_i}{\partial x_n} \frac{\partial p_m}{\partial p_n} \right] \right]$$

$$= e \left[\delta_{ik} \frac{\partial A_j}{\partial x_k} - \frac{\partial A_i}{\partial x_n} \delta_n^m \right] = e \left[\frac{\partial A_j}{\partial x_i} - \frac{\partial A_i}{\partial x_j} \right]$$

$$= e \left[\frac{\partial A_j}{\partial x_i} - \frac{\partial A_i}{\partial x_j} \right] \quad \vec{B} = \vec{\nabla} \times \vec{A}$$

$$= \epsilon^{ijk} \frac{\partial A_k}{\partial x_j} \hat{x}_i$$

$$= \epsilon^{kij} \frac{\partial A_j}{\partial x_i} \hat{x}_k \quad \text{para adequar}$$

fazendo

$$\{\pi_i, \pi_j\} \hat{x}_k = e \epsilon^{ijl} \frac{\partial A_l}{\partial x_i} \hat{x}_k$$

Pela antisimetria

$$\{x_j, \pi_i\} \hat{x}_k = -e \epsilon^{ijl} \frac{\partial A_l}{\partial x_i} \hat{x}_k$$

substituindo

$$\epsilon^{ijl} \{\pi_i, \pi_j\} \hat{x}_k = 2e \epsilon^{ijl} \frac{\partial A_l}{\partial x_i} \hat{x}_k$$

$$\frac{1}{2e} \epsilon^{ijl} \{\pi_i, \pi_j\} \hat{x}_k = \epsilon^{ijl} \frac{\partial A_l}{\partial x_i} \hat{x}_k$$

$$\rightarrow \text{Logo: } \vec{B} = \vec{\nabla} \times \vec{A} = \frac{1}{2e} \epsilon^{kij} \{\pi_i, \pi_j\} \hat{x}_k$$

$$b) \quad \dot{\vec{r}} = \{\vec{r}, H\} \quad \dot{\vec{x}} = \{\vec{x}, H\} \quad \vec{p} \cdot \vec{p} = e \vec{p} \cdot \vec{A} + e^2 \vec{A} \cdot \vec{A}$$

$$\begin{aligned} \{\vec{x}_i, H\} &= \{x_i, \frac{1}{2m} [\vec{p}_i - e \vec{A}_i]^2 + e \psi\} \quad \{x_i, \psi(x)\} = 0 \\ &= \frac{1}{2m} \left[(p_i - e A_i) \cdot \{x_i, p_j - e A_j\} + \{x_i, p_j - e A_j\} \cdot (p_i - e A_i) \right] \\ &= \frac{1}{2m} \left[2(p_j - e A_j) \{x_i, p_j\} - e \{x_i, A_j\} \right] \\ &\quad \delta_j^i \quad \downarrow A(x) \rightarrow 0 \\ &= \frac{1}{2m} \left[2(p_j - e A_j) \delta_j^i \right] = \frac{p_i - e A_i}{m} = \dot{\vec{x}} \end{aligned}$$

$$\begin{aligned} \{\vec{p}_i, H\} &= \{p_i - e A_i, \frac{1}{2m} [\vec{p}_i - e \vec{A}_i]^2 + e \psi\} \\ &= \frac{1}{2m} (p_j - e A_j) \left[2\{p_i, p_j - e A_j\} \right] - \frac{e}{2m} (p_j - e A_j) \left[2\{A_i, p_j - e A_j\} \right] + e \{p_i, \psi\} - e^2 \{A_i, \psi\} \\ &= \frac{(p_j - e A_j)}{m} \left[\cancel{\{p_i, p_j\}} - e \{p_i, A_j\} - e \{A_i, p_j\} + e^2 \{A_i, A_j\} \right] + \Omega \\ &= \frac{(p_j - e A_j)}{m} \left[-e (\{A_i, p_j\} + \{p_i, A_j\}) \right] + \Omega \\ &= \frac{(p_j - e A_j)}{m} \left[e \left(\frac{\partial A_j}{\partial x_i} - \frac{\partial A_i}{\partial x_j} \right) \right] + e \{p_i, \psi\} \\ \{p_i, \psi\} &= \frac{\partial p_i}{\partial x_i} \frac{\partial \psi}{\partial p_i} - \frac{\partial \psi}{\partial x_i} \frac{\partial p_i}{\partial p_i} = -\frac{\partial \psi}{\partial x_i} = -\nabla \psi \end{aligned}$$

$$\dot{\vec{r}} = e (\dot{\vec{x}} \times \vec{B} + \vec{E})$$

Exercício 2

sendo $L_i = \epsilon_{ijk} x_j p_k$

$$H = \frac{\vec{p}^2}{2m} - \frac{\alpha}{r} \quad r = \|\vec{r}\|, \text{ logo } \frac{\alpha}{r} \text{ é escalar}$$

e como $\vec{p}^2 = \vec{p} \cdot \vec{p}$
 $\vec{p}^2$ é escalar

i)

a) $\{L_i, H\} = \{L_i, \frac{\vec{p}^2}{2m} - \frac{\alpha}{r}\}$

$$= \frac{1}{2m} \{L_i, \vec{p}^2\} - \{L_i, \frac{\alpha}{r}\}$$

como ambos os parênteses são entre L_i e escalares

$$= \frac{1}{2m} \cdot 0 - 0 = 0$$

~~ii)~~ $\{A_i, H\}$

com $\vec{A} = \vec{p} \times \vec{L} - m\alpha \vec{r}$

$$= \{A_i, \frac{\vec{p}^2}{2m} - \frac{\alpha}{r}\} = \frac{1}{2m} \{A_i, \vec{p}^2\} - \{A_i, \frac{\alpha}{r}\}$$

Novamente, ambos parênteses envolvem escalares,
 logo são nulos

$$= 0$$

b) sendo $\{L_i, x_j\} = \epsilon_{ijk} x_k$ e $\{L_i, p_j\} = \epsilon_{ijk} p_k$

$$\{L_i, L_j\} = \{L_i, \epsilon^{jab} x_a p_b\} = \epsilon^{jab} (\{L_i, x_a\} p_b + \{L_i, p_b\} x_a)$$

$$= \cancel{\epsilon^{iab} \epsilon^{jcd} x_c p_d} \{L_i, L_j\} = \epsilon^{iab} \epsilon^{jcd} \{x_c p_d, x_a p_b\}$$

$$= \epsilon^{iab} \epsilon^{jcd} \left[p_d (p_b \{x_c, x_a\}) + x_a \{x_c, p_b\} + x_c (x_a \{p_d, p_b\}) + p_b \{p_d, x_a\} \right]$$

$$= \epsilon^{iab} \epsilon^{jcd} (p_d x_a \{x_c, p_b\} + x_c p_b \{p_d, x_a\}) = \epsilon^{iab} \epsilon^{jcd} (p_d x_a \delta_c^b - x_c p_b \delta_a^d)$$

$$= \cancel{\epsilon^{iab} \epsilon^{jcd} x_c p_b} + \epsilon^{iab} \epsilon^{jcd} x_a p_d, \quad \epsilon^{dic} \epsilon^{djh} = \delta_i^d \delta_c^h - \delta_i^h \delta_c^d$$

$$\epsilon^{cid} \epsilon^{caj} = \delta_i^c \delta_a^j - \delta_i^j \delta_a^c$$

$$= (\delta_i^c \delta_j^b - \delta_i^b \delta_j^c) x_c p_b + (\delta_i^d \delta_a^j - \delta_i^j \delta_a^d) x_a p_d$$

$$= -x_j p_i + \delta_i^j x_b p_b + x_i p_j - \delta_i^j x_d p_d$$

$$= x_i p_j - x_j p_i \quad \delta_i^j (x_b p_b - x_d p_d) = \epsilon^{ijk} L_k$$

$$\epsilon^{ijk} x_j p_i = L_k$$

$$\begin{aligned}
\{L_i, A_j\} &= \{L_i, \epsilon^{iab} p_a L_b - m \alpha x_j\} \quad x_c p_d \\
&= \epsilon^{iab} (\{L_i, p_a\} L_b + \{L_i, L_b\} p_a) - m \alpha \{L_i, x_j\} \\
&= \epsilon^{iab} \epsilon^{icd} [\{x_c, p_a\} p_d + \{p_d, p_a\} x_c] L_b + \cancel{p_a p_a} \epsilon^{iab} \epsilon^{ibc} L_c p_a - \\
&\quad - m \alpha \epsilon^{ijk} x_k \\
&= \epsilon^{iab} \epsilon^{icd} L_b [\delta_a^c p_d + 0] + \epsilon^{iab} \epsilon^{ibc} L_c p_a - m \alpha \epsilon^{ijk} x_k \\
&= \epsilon^{iab} \epsilon^{iad} L_b p_d - m \alpha \epsilon^{ijk} x_k + \epsilon^{iab} \epsilon^{ibc} L_c p_a \\
&= \delta_i^j L_b p_b - L_i p_j - m \alpha \epsilon^{ijk} x_k + L_j p_i - \delta_j^i L_c p_c
\end{aligned}$$

$$\begin{aligned}
&\epsilon^{abij} \epsilon^{adi} \\
&\delta_i^j \delta_b^d - \delta_b^i \delta_d^j \\
&\epsilon^{bja} \epsilon^{bci} \\
&\delta_a^i \delta_j^c - \delta_a^c \delta_j^i
\end{aligned}$$

$$\begin{aligned}
&= p_i L_j - p_j L_i - m \alpha \epsilon^{ijk} x_k \\
&= \epsilon^{ijk} (p_i L_j - m \alpha x_k) = \epsilon^{ijk} A_k
\end{aligned}$$

$$\begin{aligned}
\{A_i, A_j\} &= \{\epsilon^{iab} p_a L_b - m \alpha x_i, \epsilon^{jcd} p_c L_d - m \alpha x_j\} \\
&= \epsilon^{iab} \epsilon^{jcd} [L_b L_d \{p_a, p_c\} + L_b p_c \{p_a, L_d\} + p_a L_d \{L_b, p_c\} + p_a p_c \{L_b, L_d\} - \\
&\quad - m \alpha L_d \{x^i, p_c\} - m \alpha p_c \{x^i, L_d\} + m^2 \alpha^2 \{x^i, x^j\}] \\
&= \epsilon^{iab} \epsilon^{jcd} [L_b p_c (-\epsilon^{dam} p_m) + p_a L_d (\epsilon^{bcn} p_n) + p_a p_c (\epsilon^{bde} L_e) - \\
&\quad - m \alpha L_d \delta_c^i - m \alpha p_c (-\epsilon^{dip} x_p)]
\end{aligned}$$